

STEM4U — LEGO Computer Science & AI · Grades K-2 (Sparks) · 30-Lesson Plan

Course Overview

Thirty weekly 60-minute sessions built on the LEGO® Education Computer Science & AI Kit (45520), coded with **icon-only picture blocks** on LEGO's Coding Canvas — no reading or typing required, and a simple PIN gets kids in (no logins, no accounts). Groups of 4 work in pairs (Driver/Navigator). Single grade band, grades K-2 (the **Sparks**). Real tools. Real teams. Real engineers.

By the end, your child can build and code a small machine that runs steps in order, reacts when something happens, repeats a pattern, makes a simple choice, and counts — and they can explain, in their own words, how a computer learns to sort things from examples.

AI runs through every unit — honestly. Kids don't wait until Lesson 26 to meet artificial intelligence. Every lesson includes a 2-minute "AI Connection" (the **coach reads it aloud**) that ties the day's coding idea to how real AI works. At this age the promise is clear and true: **K-2 kids explore how AI works — they do not build or train their own AI model.** In the final unit they sort example pictures, play "teach the computer by showing pictures" games, and watch a coach-run demo (Google Teachable Machine) learn from examples the kids feed it. The model belongs to the demo, not the child. That honest framing is a feature, not a limitation — it leaves the real "train your own classifier" work for the 3-5 band, where it belongs.

Scope & sequence

Unit	Lessons	Big idea	What your child can do by unit's end
1 · Basics	1-5	A program is steps in order	Build and code a machine that runs steps they put in order
2 · Events	6-10	Something makes the program start	Make a model react to a press, a tap, or a sensor
3 · Loops	11-15	Computers can repeat for us	Build a machine that repeats a pattern with a loop
4 · Conditionals	16-20	Programs can make a choice	Code a model that picks one thing or another by itself
5 · Variables	21-25	Programs can count and remember	Build a model that counts and reacts when it reaches a number
6 · AI & Data	26-30	Machines can sort and learn from examples	Explore how a computer sorts pictures, and use AI recognition to drive a build

Standards & bridges: Lessons are aligned to the CSTA K-12 CS Standards — primarily Algorithms & Programming (1A-AP), with connections to Data & Analysis (1A-DA) and Impacts of Computing (1A-IC). (See the **Standards Crosswalk** at the end for the specific codes per unit.) The lessons build the earliest mental models — order, triggers, repeating, choosing, counting, and sorting — that bridge toward the 3-5 band and, later, ScratchJr and Scratch.

Rotate the "Concept + demo" format so the same 8 minutes feels fresh week to week. Pick one each lesson: **(a) straight demo** (coach shows the idea), **(b) "the silly robot"** (coach codes it wrong on purpose; kids catch the mistake and cheer the fix), **(c) "guess what it does"** (predict with a thumbs-up/down before the coach runs it), or **(d) "show a friend"** (a pair who gets it shows another pair). Keep demos short — little ones watch best in 2-minute bursts, so talk, run, react, repeat.

Every week gets a build/ownership hit. On mostly-code weeks, spend a spare minute on a **"make-it-yours" beat** — snap on one brick, give the bot a LEGO face, or name it — so no week is screen-only. Little hands need to touch bricks every session.

The six design challenges are themed so each feels like a party, with a varied show-and-tell: **L5 Build-a-Bot parade** (bots march on their trays, gallery walk) · **L10 Press-mine fair** ("make my model wake up!") · **L15 Pattern party** (everyone's machine repeats at once, like a band) · **L20 Smart-pet showcase** (two pairs show how their pets decide) · **L25 Counting carnival** (play each other's counting games) · **L30 AI story circle** (perform for a short parent-note video).

Progress layer: Each unit ends with a design-challenge lesson (5, 10, 15, 20, 25, 30). At each of these, the coach sends home a **short parent note** ("here's what your child can do now") plus a **photo of the design-challenge build**, so families can see growth across the year.

How to read a lesson (for coaches with no CS background)

Every lesson follows the same shape and the same clock, so you can teach it cold: an "I can..." goal, a 2-line AI Connection to **read aloud**, a materials note (most weeks the model is already built from earlier in the unit), a **timed 60-minute flow**, Driver/Navigator pairing, differentiation (Support / Extension / Reteach), an exit check, and an optional short at-home idea. Extensions double as **fast-finisher** tasks; the Reteach line is your quick fix if a pair is stuck. Keep your own instructions to one short sentence at a time — model it, don't explain it.

The 60-minute clock (every lesson): Warm-up (song/story/movement) **10** · Concept + demo **8** · Build/Code **25** (with a stand-up wiggle break at the halfway swap) · Challenge **7** · Share + tidy-up **10**. The clock gives little kids more movement at the start and a longer, calmer pack-up — younger focus comes in short bursts, so we front-load energy and keep hands-on time in a 25-minute window.

Model reuse: Each unit's LEGO model is built once, in the unit's first lesson, and stays assembled for the rest of the unit. Later lessons in a unit are mostly coding (with only small changes), so class time goes to thinking and playing, not rebuilding. Store each group's model on a labeled tray between sessions.

Pairing: Driver taps the picture blocks and presses Run; Navigator holds the picture-card plan and points to the next step. They **swap at the halfway mark** of Build/Code every lesson (that's the wiggle break — stand, stretch, switch seats). Odd number of kids? Make one **rotating trio** — three jobs (Driver, Navigator, Tester) that rotate at each third. Because readers and pre-readers mix at this age, pair a confident picture-reader with a partner when you can.

Lesson 1 — Basics: Meet the Kit & First Press

Objective (I can...): I can find my way around the Coding Canvas with pictures, and press two picture blocks together to make a motor move. **CS concept:** What a program is (steps a machine follows) · **Builds on:** No prior lesson — course launch.

AI Connection (2 min, read aloud): A tablet that unlocks when it sees your face is a computer following steps and making a choice. *Ask the group:* "If a robot wanted to know 'is this a cat?', what tiny things could it look at?" (Just collect ideas — we'll come back to this all year.)

Materials: LEGO CS & AI kit 45520 (1 per pair), a device per pair (Coding Canvas, PIN ready — no login), a big printed "kit parts" picture card per pair. **No full build today** — use the kit's quick-start model (hub plus one motor already on). The unit's real model is built next lesson.

60-minute flow - Warm-up (10): "Robot chef" — a child gives the coach spoken steps to make a pretend sandwich; the coach does exactly what's said, silly mistakes and all (butters the plate, etc.). Big point, lots of giggles: machines do EXACTLY what you tell them. - **Concept + demo (8):** Turn on the hub; show it connecting to the tablet. Point to the picture palette, the big green Run, and the red Stop. Drag a start picture + one "spin motor" picture; press Run so a real motor moves. Let them gasp. - **Build/Code (25):** Each pair (1) gets their hub connected with coach help, (2) takes a picture tour — find the palette, Run, Stop, (3) places 2 picture blocks (start + spin) on the quick-start model and presses Run. *Wiggle break + swap at ~12 min.* Coach circulates so every hub is connected. - **Challenge (7):** Add a "stop after a moment" picture block so the motor stops on its own. Change which picture is first and guess what happens. - **Share + tidy-up (10):** Each pair presses Run for the group and points to one kit part by its picture. Coil cables, hub off, parts in the tray. Sing a "clean-up" count-down.

Pair roles: Driver connects (with help) and drags pictures; Navigator holds the parts card and finds each part. Swap at the wiggle break. (*Trio: add a Tester who presses Run and tells the group what moved.*)

Differentiation: *Support* — a palette card so kids match blocks by picture, not words. *Extension / fast-finisher* — add a sound picture when the motor stops. *Reteach* — connect the hub together slowly; have the child name each button before pressing.

Exit check: Pair has a working hub, ran a 2-block program that moves a motor, and can point to the hub, a motor, and the green Run. **At-home (optional):** Tell a grown-up 4 exact steps to put on your shoes, like talking to a robot. ~5 min.

Lesson 2 — Basics: Build the Model & Put Steps in Order

Objective (I can...): I can build the unit model and put picture blocks in order so it does one thing after another. **CS concept:** Sequencing (steps run top to bottom, in order) · **Builds on:** Lesson 1 (running a program).

AI Connection (2 min, read aloud): When you talk to a voice helper, it follows steps in order: hear you → think → answer. *Ask the group:* "What if it tried to answer BEFORE it heard you? Would that work?"

Materials: LEGO CS & AI kit 45520 (1 per pair), device per pair, 4 big "make toast" picture cards per pair.

This is the unit's build lesson — build the Basics model now and keep it assembled through Lesson 5.

60-minute flow - Warm-up (10): Card scramble — pairs put 4 out-of-order "make toast" picture cards in the right order on the rug, then act it out. Laugh at what happens if you eat the toast before you toast it. -

Concept + demo (8): Show the same picture blocks in two different orders; run both so kids SEE different results. Say the rule together: "top to bottom, in order." - **Build/Code (25):** Pairs **build the Basics model** following the Canvas picture steps (the one build for the unit), then make a 3-4 step program (light on → motor spins → wait → stop) and run it, pointing at each step as it happens. *Wiggle break + swap at ~12 min.* -

Challenge (7): Move the same blocks into a new order that still makes sense; run and say what changed. -

Share + tidy-up (10): Each pair shows "order 1" vs. "order 2." **Leave the model built** on its tray.

Pair roles: During the build, Driver snaps bricks while Navigator points to the picture steps; during coding they swap. Swap at the wiggle break. *(Trio rotates build/point/test.)*

Differentiation: *Support* — pre-place the start block so they add the rest. *Extension / fast-finisher* — make a 5-step program. *Reteach* — touch each block and count "1, 2, 3, 4" out loud, running one step at a time.

Exit check: Pair runs an ordered program and can show why a different order gave a different result. **At-home (optional):** Draw a 4-picture comic of waking up, in order. ~5 min.

Lesson 3 — Basics: Break the Big Job into Little Steps

Objective (I can...): I can take one big action and break it into little steps before I code it. **CS concept:** Decomposition (splitting a job into smaller parts) · **Builds on:** Lesson 2 (ordered steps).

AI Connection (2 min, read aloud): To spot a cat, a computer looks at little clues — pointy ears, whiskers, a tail — then puts them together. *Ask the group:* "What little clues tell YOU it's a cat and not a dog?"

Materials: LEGO CS & AI kit + device per pair, big sticky notes or a mini picture-plan mat per pair. **Model reuse:** Basics model from Lesson 2 (no rebuild) — code plus one tiny brick tweak.

60-minute flow - Warm-up (10): "Program the human robot" — one child is the robot; the partner gives only tiny steps ("step... step... turn") to walk them to a spot. If a step is too big or fuzzy, the robot freezes with arms out. Point: machines need little, exact steps — that's breaking it down. - **Concept + demo (8):** Coach picks "make it wave hello," says the little steps out loud, then turns each into one picture block. (*Try "the silly robot" — leave out a step and let them catch it.*) - **Build/Code (25):** Using the built model, pairs put one picture-step per sticky note first, then place one block per step and run. **Make-it-yours beat:** add one brick or a LEGO face and name the bot. *Wiggle break + swap at ~12 min.* - **Challenge (7):** Add one more little step (a light or a sound) and show where it goes in the line. - **Share + tidy-up (10):** Pairs show their sticky-note line next to the running, now-named bot. Model stays built.

Pair roles: Navigator lays out the sticky-note steps; Driver codes them. Swap at the wiggle break. (*Trio: third child points to each step as it runs.*)

Differentiation: *Support* — a half-filled step line to finish. *Extension / fast-finisher* — break a two-action job into steps. *Reteach* — act the behavior out with bodies, naming each step, then map to blocks.

Exit check: Pair's sticky-note steps match the blocks in their working program. **At-home (optional):** Break "pour a bowl of cereal" into 5 little steps with a grown-up. ~5 min.

Lesson 4 — Basics: Say the Plan First (Picture Plan)

Objective (I can...): I can tell my plan in words or pictures first, then turn it into blocks. **CS concept:** Planning first / early pseudocode — saying the plan in plain words is a first taste of **picking the big steps and hiding the little mess.** · **Builds on:** Lesson 3 (little steps).

AI Connection (2 min, read aloud): Grown-ups who make AI say its plan in plain words first — "if it sees a face, wave" — before any code. *Ask the group:* "Why say your plan out loud before you build it?"

Materials: LEGO CS & AI kit + device per pair, a picture "plan mat" per pair, a stack of blank "mission cards."

Model reuse: same Basics model from Lesson 2 — code plus one tiny brick tweak.

60-minute flow - Warm-up (10): "Secret mission" — each pair draws or says a 3-step plan on a mission card, then trades cards with another pair. Your card is the other pair's mission to build and run. Clear plans win the mission. - **Concept + demo (8):** Coach says 3 plain-word lines, then matches each to a picture block while kids watch — the plan is the bridge from idea to code. (*Try "guess what it does" with thumbs up/down before Run.*) - **Build/Code (25):** Pairs build the mission card they got: say or draw the plan on the mat, then code it step-by-step on the built model and run to check it matches. **Make-it-yours beat:** add or swap one brick to fit the mission's theme. *Wiggle break + swap at ~12 min.* - **Challenge (7):** Change one plan step (like a longer wait) and fix the matching block so plan and code still agree. - **Share + tidy-up (10):** Pairs point to their plan, then run to prove it matches. Model stays built.

Pair roles: Navigator owns the plan mat; Driver turns each line into a block. Swap at the wiggle break. (*Trio: third child checks each plan step got coded.*)

Differentiation: *Support* — a plan mat with picture slots to fill ("First . **Then** ."). *Extension / fast-finisher* — a 5-step plan. *Reteach* — code one plan step together, then let them do the next alone.

Exit check: Pair's plan steps line up one-to-one with their working blocks. **At-home (optional):** Draw a 4-picture plan for getting ready for bed. ~5 min.

Lesson 5 — Basics: Design Challenge — Build-a-Bot Parade

Objective (I can...): I can plan, build, and code a little machine that does steps I chose, in order. **CS**

concept: Putting order, little steps, and a plan together · **Builds on:** Lessons 1–4. *Also serves as flex/catch-up — if earlier lessons ran long, use today to finish and celebrate.*

AI Connection (2 min, read aloud): Making an AI toy is the same loop you'll use today: break it down → plan → build → try it. *Ask the group:* "Which of those steps do you think people redo the most?"

Materials: LEGO CS & AI kit + device per pair, plan mats, a simple picture challenge card. **Model already built** (Basics model from Lesson 2) — so the 25 minutes is plan + code + play, not build-from-scratch. Pairs may add small bricks only.

60-minute flow - Warm-up (10): Review song/carousel — fast call-and-response: "What's a program?" "Steps in order!" "What's breaking it down?" "Little steps!" - **Concept + demo (8):** Show the open challenge: make the bot do 3+ steps in an order you choose. Coach says a plan for one idea, then hands off. - **Build/Code (25):** Pairs plan and code a 3+ step program on the built model (small brick changes allowed). Coach cheers and nudges, doesn't solve. *Wiggle break + swap at ~12 min.* - **Challenge (7):** Add one "surprise" step (light, sound, or extra move) and add it to the plan. - **Share + tidy-up (10): Build-a-Bot parade** — bots march on their trays in a gallery walk; each pair names the one thing that helped most. **Coach photographs each build for the unit parent note.**

Pair roles: Co-plan, then Driver codes / Navigator points to the plan; swap at the wiggle break. *(Trio rotates plan/code/test.)*

Differentiation: *Support* — a 2-step target with a partial plan to finish. *Extension / fast-finisher* — require 5 steps with a light and a sound. *Reteach* — hand them their Lesson 4 plan as a starting template.

Exit check: Pair shows a working 3+ step program with a matching plan and can tell what it does. **Progress layer:** Send home the **Unit 1 parent note** ("your child can plan and code steps in order") with the **build photo**. **At-home (optional):** Draw a dream robot and label 3 things it does in order. ~5 min.

Lesson 6 — Events: Build the Model & First Trigger

Objective (I can...): I can build the events model and make my program start when something happens. **CS concept:** Events (a trigger that starts code) · **Builds on:** Lesson 5 (programs run from a start).

AI Connection (2 min, read aloud): AI reacts to what it senses — a camera noticing your face is an "event." *Ask the group:* "What could our robot notice that would make it start?"

Materials: LEGO CS & AI kit + device per pair. **This is the unit's build lesson** — build the Events model now and keep it assembled through Lesson 10.

60-minute flow - Warm-up (10): "What made it happen?" — coach calls a trigger (doorbell rings!) and kids do the response (run to the door!). Keep it loud and moving: knock → wave, clap → jump. - **Concept + demo (8):** Show "start when I press Run" vs. "start when something happens." Demo a program that runs only when a button/tap event fires. - **Build/Code (25):** Pairs **build the Events model** (the unit's one build), then put a short sequence under an event picture (like a button press) so it acts only when triggered. *Wiggle break + swap at ~12 min.* - **Challenge (7):** Swap in a different trigger; guess, then test, how the start changes. - **Share + tidy-up (10):** Pairs show their event-triggered program and name the trigger. **Leave the model built.**

Pair roles: Driver builds/codes, Navigator points to the picture steps and names the trigger. Swap at the wiggle break. (*Trio rotates.*)

Differentiation: *Support* — a palette card showing the event blocks. *Extension / fast-finisher* — use two different triggers. *Reteach* — press the button with no event block (nothing happens), then add the event together.

Exit check: Pair's program runs only when the chosen event happens, not on its own. **At-home (optional):** Find 3 things at home that "start" something (light switch, doorbell). ~5 min.

Lesson 7 — Events: Press & Tap Triggers

Objective (I can...): I can make my model wake up from a button press or a screen tap. **CS concept:** Input events from a control · **Builds on:** Lesson 6 (events as start signals).

AI Connection (2 min, read aloud): Tapping a smart speaker or saying "hey" wakes up the helper — that's an input event. *Ask the group:* "Why might a device wait for a signal instead of always listening?"

Materials: LEGO CS & AI kit + device per pair. **Model reuse:** Events model from Lesson 6 — code only.

60-minute flow - Warm-up (10): "Human buttons" — one child is a "button," another is the "machine" that only moves when pressed on the shoulder. Giggle at pressing and nothing happening unless the button is real.

- **Concept + demo (8):** Show two event pictures (a press and a screen tap) driving two different actions. -

Build/Code (25): On the built model, pairs code two triggers — one press, one tap — each starting a different short action. *Wiggle break + swap at ~12 min.* - **Challenge (7):** Make one press do two things (move then light) while a tap does something else. - **Share + tidy-up (10):** Pairs show both triggers and say which does what. Model stays built.

Pair roles: Driver codes trigger 1, Navigator plans trigger 2; swap at the wiggle break so each builds one. (*Trio: third owns testing both.*)

Differentiation: *Support* — one trigger only, with a twin example to copy. *Extension / fast-finisher* — add a third trigger. *Reteach* — do one trigger start-to-finish together, then let them mirror it.

Exit check: Pair shows two different inputs making two different things happen. **At-home (optional):** Draw a pretend remote; draw what each button does. ~5 min.

Lesson 8 — Events: The Model Can Sense (Distance & Color)

Objective (I can...): I can make my model react when a sensor notices something. **CS concept:** Sensor-driven events · **Builds on:** Lesson 7 (press/tap inputs).

AI Connection (2 min, read aloud): A self-driving car's camera "seeing" a stop sign is a sensor event, like your color sensor spotting a card. *Ask the group:* "Who does a self-driving car help? Could it ever worry someone?"

Materials: LEGO CS & AI kit + device per pair, a small object or color card to trigger the sensor. **Model reuse:** Events model from Lesson 6; snap the sensor on as a tiny change if needed — no full rebuild.

60-minute flow - Warm-up (10): "Sensors around us" — kids name real sensors they know (doors that open by themselves, hand dryers) and act out what each one notices. - **Concept + demo (8):** Show a distance or color sensor picture firing an event when something comes near or a color shows up. - **Build/Code (25):** Pairs code the model so an action happens when the sensor notices an object/color (hand near → light on). *Wiggle break + swap at ~12 min.* - **Challenge (7):** Change how near, or which color; see how it changes. - **Share + tidy-up (10):** Pairs show the sensor making their action happen. Model stays built.

Pair roles: Driver codes, Navigator holds the trigger object and watches. Swap at the wiggle break. (*Trio rotates code/trigger/watch.*)

Differentiation: *Support* — a picture-plan slot ("WHEN sensor sees , **THEN** ") to fill. *Extension / fast-finisher* — two sensor triggers doing different things. *Reteach* — watch the live sensor picture on screen while moving the object closer.

Exit check: Pair's model reliably reacts to the sensor event, not to Run. **At-home (optional):** Find a door or light at home that senses you; tell a grown-up how it knows. ~5 min.

Lesson 9 — Events: Lots of Triggers at Once

Objective (I can...): I can set up more than one trigger so my model can react in different ways. **CS**

concept: More than one event handler · **Builds on:** Lesson 8 (sensor events).

AI Connection (2 min, read aloud): A smart home watches many things at once — the door, the temperature, your voice. *Ask the group:* "How can one helper pay attention to a few things at the same time?"

Materials: LEGO CS & AI kit + device per pair, trigger objects/color cards. **Model reuse:** Events model from Lesson 6 — code only.

60-minute flow - Warm-up (10): "Many buttons" — coach calls mixed triggers fast (knock! wave! stomp!) and kids do the matching action. Great wiggle energy. - **Concept + demo (8):** Build two separate event stacks (a tap and a sensor) in one program; show both work on their own. - **Build/Code (25):** Pairs code at least two triggers (tap → spin; object near → sound) on the built model. *Wiggle break + swap at ~12 min.* - **Challenge (7):** Add a third trigger; try them in different orders. - **Share + tidy-up (10):** Pairs show each trigger working by itself. Model stays built.

Pair roles: Each partner owns one trigger; swap at the wiggle break to try each other's. (*Trio: three triggers, one each.*)

Differentiation: *Support* — a worked twin trigger to copy the pattern from. *Extension / fast-finisher* — set off two triggers almost at once and watch. *Reteach* — build one trigger, test it, then copy and change it for the second.

Exit check: Pair's program has two+ triggers that each work when set off. **At-home (optional):** Draw a machine that reacts to 3 different things. ~5 min.

Lesson 10 — Events: Design Challenge — Wake-Up Gadget

Objective (I can...): I can design a model that reacts to at least two triggers I choose. **CS concept:** Using events for a build with a purpose · **Builds on:** Lessons 6–9. *Also serves as flex/catch-up if the unit ran long.*

AI Connection (2 min, read aloud): A smart doorbell reacts to two things — motion AND a face it knows. *Ask the group:* "For YOUR gadget, should it need both triggers, or just one?"

Materials: LEGO CS & AI kit + device per pair, plan mats, trigger objects. **Model already built** (Events model from Lesson 6) — the 25 min is plan + code + play, small changes only.

60-minute flow - Warm-up (10): Rapid review — pairs name every trigger they used (press, tap, near, color) and act one out. - **Concept + demo (8):** Challenge: build a "wake-up gadget" or "greeter" that reacts to two triggers. Coach says a plan for one trigger, then steps back. - **Build/Code (25):** Pairs plan and code two trigger-reactions on the built model (motion near → alarm sound; tap → reset light). *Wiggle break + swap at ~12 min.* - **Challenge (7):** Add a third trigger or make one reaction do two things. - **Share + tidy-up (10): Press-mine fair** — before a pair sets off their gadget, the audience guesses which triggers it reacts to, then the pair sets them off live to check. **Photograph each build.**

Pair roles: Co-plan; Driver codes / Navigator points to the plan; swap at the wiggle break. *(Trio rotates plan/code/test.)*

Differentiation: *Support* — a one-trigger target with a partial plan. *Extension / fast-finisher* — require a sensor trigger AND a control trigger. *Reteach* — reuse a Lesson 7–9 program as a starting template.

Exit check: Pair shows a working model with two+ different trigger-reactions and a matching plan. **Progress layer:** Send home the **Unit 2 parent note** ("your child can make a model sense and react") with the **build photo**. **At-home (optional):** Invent an alarm for something at home; name its triggers. ~5 min.

Lesson 11 — Loops: Build the Model & Do It Again

Objective (I can...): I can build the loops model and use a repeat block so I don't copy the same block many times. **CS concept:** Loops / repeating (repeat N times). A loop is **hiding the mess**: one "repeat 5x" picture stands in for five copied blocks. · **Builds on:** Lesson 10 (events start actions we can now repeat).

AI Connection (2 min, read aloud): AI gets better by repeating — it sees examples over and over to learn. *Ask the group:* "Why would doing it many times help a computer learn?"

Materials: LEGO CS & AI kit + device per pair. **This is the unit's build lesson** — build the Loops model now and keep it assembled through Lesson 15.

60-minute flow - Warm-up (10): "Do it again" — clap 10 times counting together, then find a shorter way to say it: "repeat 10 times!" Sing a repeating song (like a 2-line chorus over and over) and name the repeat. -

Concept + demo (8): Show 4 copied "move" blocks, then replace them with one move inside a "repeat 4" picture; run both so kids see they match. - **Build/Code (25):** Pairs **build the Loops model** (the unit's one build), then code a repeated action with a repeat picture (spin, pause, repeat 5x). *Wiggle break + swap at ~12 min.* - **Challenge (7):** Change the repeat number; guess how much longer it runs before you press Run. -

Share + tidy-up (10): Pairs show their loop and say the count out loud. **Leave the model built.**

Pair roles: Driver builds/codes, Navigator counts the repeats out loud. Swap at the wiggle break. (*Trio rotates.*)

Differentiation: *Support* — a palette card highlighting the repeat picture. *Extension / fast-finisher* — put two blocks inside the loop. *Reteach* — count the repeats out loud together on the first run.

Exit check: Pair swaps copied blocks for a working repeat loop and can say the count. **At-home (optional):** Find something you do over and over; say it as "repeat ___ times." ~5 min.

Lesson 12 — Loops: Loops with Moves & Sounds (+ Fix-It)

Objective (I can...): I can put a few actions inside one loop so a whole pattern repeats. **CS concept:** A loop body with more than one action · **Builds on:** Lesson 11 (single-action repeat).

AI Connection (2 min, read aloud): Each training "round" repeats the same steps on new examples until the AI gets better. *Ask the group:* "What might get a little better each time the loop runs again?"

Materials: LEGO CS & AI kit + device per pair. **Model reuse:** Loops model from Lesson 11 — code only.

60-minute flow - Warm-up (10): Pattern clap — coach does "clap-stomp-clap-stomp"; kids find the chunk that repeats and do it as a group. That chunk goes in the loop. - **Concept + demo (8):** Build a loop with move + sound + wait; run it and point out the whole chunk repeats. - **Build/Code (25):** Pairs code a 3-action repeating pattern inside one loop (spin → beep → wait). **Fix-It mini-task (3 min):** hand each pair a "silly" program where a block sits OUTSIDE the loop — they find it and drag it in so the whole pattern repeats. *Wiggle break + swap at ~12 min.* - **Challenge (7):** Add a light or a backward move to the loop; run it again. - **Share + tidy-up (10):** Pairs show the pattern and name the actions inside the loop. Model stays built.

Pair roles: Driver codes the loop, Navigator names the actions to include. Swap at the wiggle break. (*Trio: third leads the Fix-It hunt.*)

Differentiation: *Support* — a 2-action loop with a twin example. *Extension / fast-finisher* — a 4-action loop. *Reteach* — build the actions outside a loop first, confirm they work, then drag them inside together.

Exit check: Pair runs a loop whose body has 3+ actions that repeat together. **At-home (optional):** Make up a 3-move dance; repeat it 4 times for a grown-up. ~5 min.

Lesson 13 — Loops: Forever & Stop

Objective (I can...): I can use a forever loop and know how to stop it. **CS concept:** Loops that run forever and how to stop them · **Builds on:** Lesson 12 (loop bodies).

AI Connection (2 min, read aloud): An always-on helper runs a forever loop, listening until you talk. *Ask the group:* "Who does an always-listening speaker help? Could it bother anyone at home?"

Materials: LEGO CS & AI kit + device per pair. **Model reuse:** Loops model from Lesson 11 — code only.

60-minute flow - Warm-up (10): "Never-ending song" — start a chant with no end and keep going; ask the kids how to make it stop. Introduce a loop that runs until we say stop. - **Concept + demo (8):** Demo a forever loop (a blinking light); stop it with the red Stop, then better, with an event. - **Build/Code (25):** Pairs code a forever loop (blink or spin) on the built model, then add a way to stop it (Stop button or an event). *Wiggle break + swap at ~12 min.* - **Challenge (7):** Use one trigger to start the forever loop and another to stop it. - **Share + tidy-up (10):** Pairs show starting and stopping. Model stays built.

Pair roles: Driver codes the loop, Navigator owns the stop plan. Swap at the wiggle break. (*Trio rotates start/loop/stop.*)

Differentiation: *Support* — use the red Stop only. *Extension / fast-finisher* — start and stop entirely with triggers. *Reteach* — run the forever loop, feel that it never ends, then add the stop together.

Exit check: Pair runs a forever loop and stops it on purpose. **At-home (optional):** Name a machine that runs until it's turned off; how would you stop it? ~5 min.

Lesson 14 — Loops: A Loop Inside a Loop (+ Fix-It)

Objective (I can...): I can put one loop inside another to make a bigger repeating pattern. **CS concept:** Nested loops · **Builds on:** Lesson 13 (loops and stopping).

Stretch, not required. A loop inside a loop is a big-kid stretch — tricky for the youngest Sparks. A single working loop (L11-13) is the real takeaway; treat nesting as a fun bonus and don't hold a struggling pair to it.

AI Connection (2 min, read aloud): AI's practice loops stack up — for each round, it loops through every example. *Ask the group:* "If we beep 3 times, and do THAT 2 times, how many beeps is that?"

Materials: LEGO CS & AI kit + device per pair, big dot cards per pair. **Model reuse:** Loops model from Lesson 11 — code only.

60-minute flow - Warm-up (10): "Rows of beats" — pairs lay out 2 rows of 3 dots and count the total by groups. Clap it: "1-2-3, again, 1-2-3." - **Concept + demo (8):** Build an inner loop (beep 3×) inside an outer loop (repeat 2×); run it so kids hear 2 sets of 3. - **Build/Code (25):** Pairs code a loop-in-a-loop (spin-beep 3×, all repeated 2×). **Fix-It mini-task (3 min):** give a program with the inner and outer numbers swapped — pairs figure out why the count is wrong and fix it. *Wiggle break + swap at ~12 min.* - **Challenge (7):** Change the inner or outer number; guess the new total before running. - **Share + tidy-up (10):** Pairs show it and say how many inner runs. Model stays built.

Pair roles: Driver builds the inner loop, Navigator adds the outer and counts. Swap at the wiggle break. (*Trio: third leads the Fix-It.*)

Differentiation: *Support* — a twin nested example to copy. *Extension / fast-finisher* — add a light inside the inner loop. *Reteach* — build and test the inner loop alone, then wrap the outer around it together.

Exit check: Pair runs a working loop-in-a-loop and can point to the inner vs. outer count. (*Stretch — not required for all.*) **At-home (optional):** Draw 2 rows of 3 stickers; say it as "a loop inside a loop." ~5 min.

Lesson 15 — Loops: Design Challenge — Pattern Party

Objective (I can...): I can design a machine that repeats a pattern I made up, using loops. **CS concept:** Using loops for a creative build · **Builds on:** Lessons 11-14. *Also serves as flex/catch-up if the unit ran long.*

AI Connection (2 min, read aloud): Repeating is how AI practices — lots and lots of loops to get one skill right. *Ask the group:* "Would your machine get better if it practiced its pattern more, like AI does?"

Materials: LEGO CS & AI kit + device per pair (with camera), plan mats. **Model already built** (Loops model from Lesson 11) — build time is plan + code + play, small changes only.

60-minute flow *(As the flex lesson, today's clock trims Build/Code to ~18 to make room for a short AI peek. If a pair needs the full build time, run the AI peek as a whole-group demo instead.)*

- **Warm-up (10):** Loop review — pairs sort example programs into "repeat N," "forever," and "loop-in-a-loop." *(Theme: Pattern party — everyone runs at once at the end.)* - **Concept + demo (8):** Challenge: build a "pattern machine" (dancer, drummer, light show) using at least one loop. Coach says a plan with a loop, then hands off. *(Try "show a friend.")* - **Build/Code (~18):** Pairs plan and code a repeating pattern that must use a loop on the built model. *Swap at the halfway mark.* - **Peek at a real AI — watch it sort (10):** The coach opens the kit's recognition feature (or Google Teachable Machine on the coach's screen) and holds up cards, small objects, or hand shapes to the camera; the whole group watches the on-screen label change and calls out the guess. **This is a watch-and-feed demo — the kids are exploring how a computer sorts, not training their own AI.** *This previews Unit 6.* **Privacy note (say it aloud):** the camera runs on the device, with no logins, and pictures are not saved. - **Share + tidy-up (10):** Pattern party — pairs run their pattern together like a band, and each points to the loop in their program. **Photograph each build.**

Pair roles: Co-plan; Driver codes / Navigator counts the loop; swap at the halfway mark. *(Trio rotates plan/code/test.)*

Differentiation: *Support* — a single-repeat target with a partial plan. *Extension / fast-finisher* — add a loop-in-a-loop plus a trigger. *Reteach* — reuse a Lesson 12-14 loop as a template.

Exit check: Pair shows a working pattern that uses a loop, and can say why one repeat block is easier than copying the blocks many times (**hiding the mess**). **Progress layer:** Send home the **Unit 3 parent note** ("your child can make a machine repeat with loops") with the **build photo**. **At-home (optional):** Make up a clap pattern; say how many times each part repeats. ~5 min.

Lesson 16 — Conditionals: Build the Model & If / Then

Objective (I can...): I can build the conditionals model and use an if/then picture so it decides what to do. **CS**

concept: Conditionals (if something is true, then act) · **Builds on:** Lesson 15 (events + loops set up decisions).

AI Connection (2 min, read aloud): AI makes choices: "IF I'm pretty sure it's a dog, THEN bark." *Ask the group:* "What should it do if it's only a tiny bit sure?"

Materials: LEGO CS & AI kit + device per pair, trigger object/color card. **This is the unit's build lesson** — build the Conditionals model now and keep it assembled through Lesson 20.

60-minute flow - Warm-up (10): "If... then..." chant — coach says a condition ("if it's raining...") and kids act the action ("...grab an umbrella!"). Big movement version: "if I say freeze, then freeze!" - **Concept + demo (8):** Demo an if/then: if the sensor sees an object, then light on; if not, nothing. - **Build/Code (25):** Pairs **build the Conditionals model** (the unit's one build), then code one if/then (object near / color seen → action). *Wiggle break + swap at ~12 min.* - **Challenge (7):** Change the condition; check the action only happens when it's true. - **Share + tidy-up (10):** Pairs show the "it happened" case and the "nothing happened" case. **Leave the model built.**

Pair roles: Driver builds/codes, Navigator says the condition out loud before each run. Swap at the wiggle break. (*Trio rotates.*)

Differentiation: *Support* — a palette card for the if/then picture. *Extension / fast-finisher* — make the "then" do two things. *Reteach* — hold the object near/far and say "true... not true..." while it runs.

Exit check: Pair's model acts only when the condition is true. **At-home (optional):** Say three "if... then..." rules for your house to a grown-up. ~5 min.

Lesson 17 — Conditionals: True or Not True? (+ Fix-It)

Objective (I can...): I can tell if something is true or not true and guess what my program will do. **CS**

concept: True/false conditions · **Builds on:** Lesson 16 (if/then).

AI Connection (2 min, read aloud): Deep down, an AI's "is this a cat?" turns into a true-or-not-true answer after it checks how sure it is. *Ask the group:* "How sure should it be before it says 'true'?"

Materials: LEGO CS & AI kit + device per pair, happy/sad or ✓/✗ cards. **Model reuse:** Conditionals model from Lesson 16 — code only.

60-minute flow - Warm-up (10): True/not-true flash — coach reads simple statements ("The hub is on," "It's snowing inside"); kids hold up ✓ or ✗ and cheer. - **Concept + demo (8):** Show the same if/then; before each run ask "is it true right now?" then run to check. - **Build/Code (25):** Pairs run their if/then once when it's true and once when it's not, guessing first each time. **Fix-It mini-task (3 min):** give a program with the condition set backwards (acts when it shouldn't) — pairs find and fix it. *Wiggle break + swap at ~12 min.* - **Challenge (7):** Add a light that turns on to show "this is true right now." - **Share + tidy-up (10):** Pairs guess, then show, both a true and a not-true result. Model stays built.

Pair roles: Driver runs and codes, Navigator guesses true/not-true before each run. Swap at the wiggle break. (*Trio: third leads the Fix-It.*)

Differentiation: *Support* — a picture checklist for reading the condition. *Extension / fast-finisher* — use a sensor amount as the condition. *Reteach* — watch the live sensor picture and say "true/not true" as the object moves.

Exit check: Pair correctly guesses and shows the action for both true and not-true. **At-home (optional):** Say 5 things about your day; tell a grown-up which are true. ~5 min.

Lesson 18 — Conditionals: If / Then / Else (This OR That)

Objective (I can...): I can use if/then/else so my model does one thing OR another. **CS concept:** Picking between two choices (else) · **Builds on:** Lesson 17 (true/not-true).

AI Connection (2 min, read aloud): AI picks a path: IF it knows your face, THEN unlock; ELSE stay locked.

Ask the group: "Face-unlock is fast — who does it help, and could it ever lock out the wrong person?"

Materials: LEGO CS & AI kit + device per pair, trigger object/color card. **Model reuse:** Conditionals model from Lesson 16 — code only.

60-minute flow - Warm-up (10): "Two doors" — coach says a condition with two outcomes ("if the light is green, go; else, stop"); kids act out both. March to green, freeze to red. - **Concept + demo (8):** Demo if/then/else: object near → green light; else → red light. Run both ways. - **Build/Code (25):** Pairs code an if/then/else with two clearly different outcomes from one condition. *Wiggle break + swap at ~12 min.* - **Challenge (7):** Make each path do two things (light + sound). - **Share + tidy-up (10):** Pairs show both paths and say which condition chose which. Model stays built.

Pair roles: Driver codes the "then," Navigator plans the "else"; swap at the wiggle break so each owns a path. (*Trio: third owns testing both.*)

Differentiation: *Support* — the "then" filled in; kids add only the "else." *Extension / fast-finisher* — put an if inside one path. *Reteach* — set off each path by hand and name it before coding.

Exit check: Pair's model shows two different outcomes picked by one condition. **At-home (optional):** Make an if/then/else rule for choosing what to wear. ~5 min.

Lesson 19 — Conditionals: The Sensor Decides (+ Fix-It)

Objective (I can...): I can use a sensor amount in a condition so my model decides by itself. **CS concept:** Conditions that use a sensor amount (a cutoff) · **Builds on:** Lesson 18 (if/then/else).

AI Connection (2 min, read aloud): AI picks a cutoff too — "IF I'm more than pretty-sure, THEN act." *Ask the group:* "What goes wrong if the cutoff is way too low?"

Materials: LEGO CS & AI kit + device per pair, objects at near/far spots, color cards. **Model reuse:** Conditionals model from Lesson 16 — code only.

60-minute flow - Warm-up (10): "How close is close?" — pairs hold hands apart to show "near" and "far," then pick a line on the rug for the cutoff and step over it. - **Concept + demo (8):** Demo a condition using a sensor amount: if closer than a set line, then act; else the other path. - **Build/Code (25):** Pairs code a sensor-cutoff if/then/else (closer than X → back away; else → wait). **Fix-It mini-task (3 min):** give a program with the cutoff set so wrong it never acts — pairs find and fix the number. *Wiggle break + swap at ~12 min.* - **Challenge (7):** Change the cutoff; test how the choice changes. - **Share + tidy-up (10):** Pairs show the model deciding differently at near and far. Model stays built.

Pair roles: Driver codes, Navigator moves the object and calls near/far. Swap at the wiggle break. *(Trio: third leads the Fix-It.)*

Differentiation: *Support* — give the cutoff number. *Extension / fast-finisher* — two cutoffs for three outcomes. *Reteach* — watch the live distance number and pick the cutoff together.

Exit check: Pair's model changes what it does when the sensor crosses the cutoff. **At-home (optional):** Find an auto light or auto door; tell a grown-up its "if/then/else" rule. ~5 min.

Lesson 20 — Conditionals: Design Challenge — Smart Pet

Objective (I can...): I can design a model that decides for itself using a condition. **CS concept:** Using if/then/else with a sensor · **Builds on:** Lessons 16–19. *Also serves as flex/catch-up if the unit ran long.*

AI Connection (2 min, read aloud): A smart doorbell decides: IF it sees a person, THEN ring; ELSE ignore. *Ask the group:* "What sensor and rule would make YOUR pet smart?"

Materials: LEGO CS & AI kit + device per pair, plan mats, trigger objects. **Model already built** (Conditionals model from Lesson 16) — the 25 min is plan + code + play, small changes only.

60-minute flow - Warm-up (10): Conditionals review — pairs match little scenarios to if/then vs. if/then/else, with actions. (*Theme: Smart-pet showcase.*) - **Concept + demo (8):** Challenge: build a "smart pet" or "helper" that picks between two actions using a sensor. Coach says a plan with a condition, then hands off. (*Try "the silly robot."*) - **Build/Code (25):** Pairs plan and code an if/then/else decision driven by a sensor on the built model. *Wiggle break + swap at ~12 min.* - **Challenge (7):** Add a second condition, or add a loop so the pet decides over and over. - **Share + tidy-up (10): Two ways to do it** — pair up two teams that met the same goal differently. Each shows theirs, then together they decide which is simpler and say why (there's often more than one right answer). **Photograph each build.**

Pair roles: Co-plan; Driver codes / Navigator checks the condition; swap at the wiggle break. (*Trio rotates plan/code/test.*)

Differentiation: *Support* — a one-condition target with a partial plan. *Extension / fast-finisher* — require a loop plus if/then/else. *Reteach* — reuse a Lesson 18–19 program as a template.

Exit check: Pair shows a model that picks between actions correctly using a condition. **Progress layer:** Send home the **Unit 4 parent note** ("your child can make a model decide on its own") with the **build photo**. **At-home (optional):** Invent a smart pet; say its one main if/then/else rule. ~5 min.

Lesson 21 — Variables: Build the Model & a Box That Holds a Number

Objective (I can...): I can build the variables model, make a "box," and put a number in it. **CS concept:** Variables (a named box that holds a number) · **Builds on:** Lesson 20 (decisions we can base on a stored number).

AI Connection (2 min, read aloud): AI remembers things as numbers — like how many examples it has seen. *Ask the group:* "What might a robot need to remember to get better at a job?"

Materials: LEGO CS & AI kit + device per pair, a real labeled paper "box" per pair with number cards. **This is the unit's build lesson** — build the Variables model now and keep it assembled through Lesson 25.

60-minute flow - Warm-up (10): "Name the box" — coach holds a labeled box ("SCORE"), drops in a number card, swaps it for a bigger one. A variable is a box with a name, and the number inside can change. -

Concept + demo (8): Make a variable "score," set it to 0, show it big on screen, then change it and watch the number update. - **Build/Code (25):** Pairs **build the Variables model** (the unit's one build), then make a variable, put a starting number in it, and show it (screen or lights). *Wiggle break + swap at ~12 min.* -

Challenge (7): Put a different starting number in; run to see the display change. - **Share + tidy-up (10):** Pairs show their named box and the number inside. **Leave the model built.**

Pair roles: Driver builds/codes, Navigator names the box and reads the number. Swap at the wiggle break. (*Trio rotates.*)

Differentiation: *Support* — a twin example showing a named box to copy. *Extension / fast-finisher* — make two boxes. *Reteach* — write a number in the real paper box, then make the screen match.

Exit check: Pair makes a named variable and puts a number in it that shows on screen. **At-home (optional):** Label 3 boxes at home ("forks," "cups") and write how many are in each. ~5 min.

Lesson 22 — Variables: Counting Up

Objective (I can...): I can change the number in my box to count things. **CS concept:** Adding to a variable (counting up) · **Builds on:** Lesson 21 (making/storing a variable).

AI Connection (2 min, read aloud): AI keeps a tally — how many examples it saw, how many it got right.

Ask the group: "What would a robot count to know it's getting better?"

Materials: LEGO CS & AI kit + device per pair, trigger object/button. **Model reuse:** Variables model from Lesson 21 — code only.

60-minute flow - Warm-up (10): Tally clap — coach makes a tally mark per clap on the board; count up together. Connect it to "add 1 to the box." - **Concept + demo (8):** Demo "add 1 to score" inside a trigger so each press adds one; show the number climbing. - **Build/Code (25):** Pairs code a counter: each trigger (press or sensor) adds 1 and shows the number. *Wiggle break + swap at ~12 min.* - **Challenge (7):** Add a reset trigger that puts the number back to 0. - **Share + tidy-up (10):** Pairs show counting up and resetting. Model stays built.

Pair roles: Driver codes the counter, Navigator sets off the trigger and reads the number. Swap at the wiggle break. (*Trio rotates code/trigger/read.*)

Differentiation: *Support* — the trigger is pre-built; kids add the "add 1." *Extension / fast-finisher* — count by 2, or keep two counters. *Reteach* — press once, watch 0 → 1, then let them guess the next.

Exit check: Pair's number goes up correctly on each trigger and can reset. **At-home (optional):** Count something for one minute (jumps, claps); write the number. ~5 min.

Lesson 23 — Variables: The Number Makes a Choice

Objective (I can...): I can use the number in my box inside an if/then so my program reacts. **CS concept:** Variables plus conditionals · **Builds on:** Lesson 22 (counting) and Lesson 18 (if/then/else).

AI Connection (2 min, read aloud): AI acts on what it remembers: IF it got enough right, THEN it's "ready."
Ask the group: "What number would tell a robot it's ready?"

Materials: LEGO CS & AI kit + device per pair, trigger object/button. **Model reuse:** Variables model from Lesson 21 — code only.

60-minute flow - Warm-up (10): "When we get to..." — coach counts claps with a rule ("at 5, everybody cheer!"). Kids feel the number trigger a reaction. - **Concept + demo (8):** Demo: count presses; if the number reaches a target, then celebrate (light + sound); else keep counting. - **Build/Code (25):** Pairs code a counter plus an if/then that does a special action when the number hits a chosen value. *Wiggle break + swap at ~12 min.* - **Challenge (7):** Change the target number, or add an "else" for "not yet." - **Share + tidy-up (10):** Pairs show reaching the target and the model reacting. Model stays built.

Pair roles: Driver codes the if/then, Navigator owns the counter and counts to the target. Swap at the wiggle break. (*Trio rotates.*)

Differentiation: *Support* — the counter is pre-built; kids add the if/then. *Extension / fast-finisher* — two targets with two reactions. *Reteach* — set the target to 2 so it triggers fast, then raise it.

Exit check: Pair's model changes what it does when the number reaches the target. **At-home (optional):** Make a rule like "when I read to 10 minutes, I get a sticker." ~5 min.

Lesson 24 — Variables: Two Boxes at Once

Objective (I can...): I can use two boxes to keep track of two things at the same time. **CS concept:** More than one variable · **Builds on:** Lesson 23 (a number in a decision).

AI Connection (2 min, read aloud): AI keeps track of many numbers at once — how fast, how many right, how many of each kind. *Ask the group:* "Why can't just one number tell a robot everything?"

Materials: LEGO CS & AI kit + device per pair, trigger objects/buttons. **Model reuse:** Variables model from Lesson 21 — code only.

60-minute flow - Warm-up (10): "Two scoreboards" — pairs keep two tally columns (points AND stars), feeling the need for two boxes. - **Concept + demo (8):** Demo two variables (a score and a count), changing each and showing both on screen. - **Build/Code (25):** Pairs code two boxes — a score that counts up on a trigger and a second count that changes in a loop — both shown. *Wiggle break + swap at ~12 min.* -

Challenge (7): Add an if/then that reacts when either number hits a value (win / done). - **Share + tidy-up (10):** Pairs show both numbers changing and say what each one tracks. Model stays built.

Pair roles: Each partner owns one box; swap at the wiggle break to try each other's. (*Trio: third owns the win/done check.*)

Differentiation: *Support* — one box pre-built; kids add the second. *Extension / fast-finisher* — end the program on a two-box rule. *Reteach* — build and test one box fully, then copy the pattern for the second.

Exit check: Pair runs a program with two working, clearly named boxes. **At-home (optional):** Think of a game; name the two things it keeps track of. ~5 min.

Lesson 25 — Variables: Design Challenge — Counting Carnival

Objective (I can...): I can design a little game or gadget that counts and reacts. **CS concept:** Using variables with events, loops, and conditionals · **Builds on:** Lessons 21–24. *Also serves as flex/catch-up if the unit ran long.*

AI Connection (2 min, read aloud): A game's AI keeps track of your score and changes to match it — the number guides its choices. *Ask the group:* "A game that changes to fit you is fun — could it ever bug a player?"

Materials: LEGO CS & AI kit + device per pair, plan mats, trigger objects. **Model already built** (Variables model from Lesson 21) — the 25 min is plan + code + play, small changes only.

60-minute flow - Warm-up (10): Variables review — "What's a variable?" "A box with a number!" "How do we count?" "Add 1!" Do it as a chant with motions. - **Concept + demo (8):** Challenge: build a game/gadget that keeps a count and reacts when it hits a target. Coach says a plan with one box, then hands off. - **Build/Code (25):** Pairs plan and code a program using at least one box plus a trigger and a reaction (a win or a milestone) on the built model. *Wiggle break + swap at ~12 min.* - **Challenge (7):** Add a second box, a reset, or a count-down. - **Share + tidy-up (10): Counting carnival** — pairs rotate to play each other's counting games, then each creator points to the box behind their scoreboard. **Photograph each build.**

Pair roles: Co-plan; Driver codes / Navigator checks the counting; swap at the wiggle break. *(Trio rotates plan/code/test.)*

Differentiation: *Support* — a count-only target with a partial plan. *Extension / fast-finisher* — require two boxes and a win rule. *Reteach* — reuse a Lesson 23–24 program as a template.

Exit check: Pair shows a working game/gadget that counts, stores a number, and reacts when it hits a value.

Progress layer: Send home the **Unit 5 parent note** ("your child can build a game that counts and reacts") with the **build photo**. **At-home (optional):** Draw your game's scoreboard; label what it counts. ~5 min.

Lesson 26 — AI & Data: How Computers Sort (Build the Model)

Objective (I can...): I can sort pictures into groups by a rule, and explain that computers can sort too. **CS concept:** Classifying / sorting by features — the first idea behind how AI recognizes things · **Builds on:** Lesson 25 (we've counted; now we group). *Honest framing: today kids sort examples themselves and watch a computer sort — no AI is trained.*

AI Connection (2 min, read aloud): AI that "knows" a cat really learned to SORT pictures into cat and not-cat. *Ask the group:* "This computer learned from lots of examples — what do you think it looked at to tell them apart?"

Materials: LEGO CS & AI kit + device per pair, a big set of picture/object cards to sort. **This is the AI unit's build lesson** — build the AI & Data model now and keep it assembled through Lesson 29.

60-minute flow - Warm-up (10): "Sort it!" — pairs sort a pile of picture cards into two groups (like animals vs. not-animals) and say their rule. Then re-sort by a new rule (big vs. small). Point: a rule tells you which group. - **Concept + demo (8):** The **coach** shows the device recognizing a simple category (or runs the Teachable Machine demo from the coach's screen) and sorting cards into groups live; the group calls out the guess. Say aloud: the computer is sorting, just like we did. On-device, saves nothing. - **Build/Code (25):** Pairs **build the AI & Data model** per the Canvas picture steps. Then they keep sorting their own cards into labeled trays and tally how many land in each group (reusing counting from Unit 5). *Wiggle break + swap at ~12 min.* - **Challenge (7):** Make a trickier card that's hard to sort; talk about why it's hard. - **Share + tidy-up (10):** Pairs show their two groups and say their sorting rule. **Leave the model built.**

Pair roles: Driver builds, Navigator sorts cards and tallies each group. Swap at the wiggle break. *(Trio rotates build/sort/tally.)*

Differentiation: *Support* — two very different card sets (easy to tell apart). *Extension / fast-finisher* — sort into three groups. *Reteach* — sort just 4 cards together, naming the rule each time.

Exit check: Pair sorts pictures into groups by a clear rule and can say that computers sort the same way. **At-home (optional):** Sort your socks or toys into 2 groups; tell a grown-up your rule. ~5 min.

Lesson 27 — AI & Data: Teach the Computer by Showing Pictures

Objective (I can...): I can give the computer example pictures and watch it learn to tell groups apart. **CS concept:** Examples are data — more clear examples help a computer sort better · **Builds on:** Lesson 26 (sorting by a rule). *Honest framing: the coach runs the model; kids feed it examples and watch — they are exploring how it learns, not training their own AI.*

AI Connection (2 min, read aloud): This is the big secret of AI: we don't write every rule — we SHOW it examples until it sorts on its own. *Ask the group:* "How many pictures of each thing do you think it needs to tell them apart?"

Materials: LEGO CS & AI kit + device per pair, the AI & Data model (from L26), a big screen for the coach's Teachable Machine demo, picture/object cards. **Model reuse:** AI & Data model from Lesson 26 — mostly watch + code.

60-minute flow - Warm-up (10): "Teach a friend" — one child describes a thing in words; the partner guesses. More clear examples make guessing easier — same for a computer. - **Concept + demo (10):** The **coach** runs Google Teachable Machine on the coach's screen as a group demo. Kids take turns holding example cards/objects to the camera to "show the computer," category by category, while the coach clicks capture. Then the group tests it and cheers the right guesses. **Say it plainly: the coach's computer is learning from the examples WE show it — we are teaching it by example, and exploring how it works. (The kids do not build their own AI.)** Restate: on-device, no logins, pictures are not saved. - **Build/Code (22):** Back at their own model, pairs connect the kit's built-in recognition so that when it recognizes a shown card, the model does an action. They keep a tally of "how many examples per group" on the plan mat and ask "are our groups even?" *Wiggle break + swap at ~12 min.* - **Challenge (7):** Add a second recognized card that makes a different action. - **Share + tidy-up (10):** Pairs show a card and the model reacting, and say "we taught it by showing examples." Model stays built.

Pair roles: Driver codes the action, Navigator holds cards to the camera and reads the label. Swap at the wiggle break. *(Trio: third keeps the example tally.)*

Differentiation: *Support* — one recognition-to-action link pre-built. *Extension / fast-finisher* — link two recognitions to two actions. *Reteach* — watch the on-screen label change as cards are shown before wiring the action.

Exit check: Pair can show the model reacting to a recognized picture and can say that computers learn to sort from examples we show them. **At-home (optional):** Show a grown-up 3 example pictures of one thing; explain why examples help a computer learn. ~5 min.

Lesson 28 — AI & Data: The Recognizer Drives the Robot

Objective (I can...): I can use what the recognizer sees to make my model move or react. **CS concept:** A recognized picture is a trigger (AI recognition as an event) · **Builds on:** Lesson 27 (showing examples) and Unit 2 events. *Honest framing: kids use a ready-made recognizer — they do not train their own AI. Privacy: on-device camera, no logins, pictures never saved.*

AI Connection (2 min, read aloud): When the camera recognizes something, that can START the robot — just like the triggers we learned, but now the trigger is "the computer sees it." *Ask the group:* "What should the robot do the moment it sees our special card?"

Materials: LEGO CS & AI kit + device per pair with camera, recognition cards/objects. **Model reuse:** AI & Data model from Lesson 26 — code only.

60-minute flow - Warm-up (10): "When you see it, go" — coach holds up a card; when kids see the special one, they do a move. Recognize → act. - **Concept + demo (8):** Demo the recognizer: when it recognizes a category, an event fires and the model moves (through a little maze or path). Note aloud: on-device, saves nothing. - **Build/Code (25):** Pairs code events so that when the recognizer sees a category, the model does an action (move, light, sound). *Wiggle break + swap at ~12 min.* - **Challenge (7):** Add a second recognized category that triggers a different move. - **Share + tidy-up (10):** Pairs show a card driving the model and name the trigger. Model stays built.

Pair roles: Driver codes, Navigator holds cards to the camera and confirms the recognition. Swap at the wiggle break. *(Trio rotates code/show/test.)*

Differentiation: *Support* — one recognition-to-move link pre-built to extend. *Extension / fast-finisher* — chain two recognitions into a path. *Reteach* — watch the label change as cards are shown before wiring the move.

Exit check: Pair's model reacts correctly when the recognizer sees the target card. **At-home (optional):** Name 3 ways phones or tablets "recognize" things (faces, voices). ~5 min.

Lesson 29 — AI & Data: Why Computers Sometimes Guess Wrong

Objective (I can...): I can tell one reason a computer might guess wrong, and one way better examples would help. **CS concept:** Example quality and fairness — more and more varied examples make sorting better · **Builds on:** Lessons 27–28. *Honest framing: kids watch the coach add examples and improve a demo model; they do not train their own AI. Privacy: on-device camera, no logins, pictures never saved.*

AI Connection (2 min, read aloud): This is why we said "more tries help" back in loops — better, more different examples help a computer sort right. *Ask the group:* "What kind of example would fix a computer that keeps guessing wrong?"

Materials: LEGO CS & AI kit + device per pair with camera, a big screen for the coach demo. **Model reuse:** AI & Data model from Lesson 26 — code and observe.

60-minute flow - Warm-up (10): "Tricky twins" — coach shows two things that look alike; ask how you'd tell them apart for sure. Connect to clearer, more varied examples. - **Concept + demo (10):** On the **coach's** screen, run the demo so it guesses wrong on purpose (maybe it only saw one kind of example). As a group, kids feed it better, more varied examples and watch the guess improve. **Say it plainly: we're exploring why it got better — more examples, and more KINDS of examples.** - **Build/Code (22):** Pairs test their own recognizer-driven model from L28, note when it guesses wrong, and (where the kit allows) show it more or clearer cards; they keep the example tally and re-check "are our groups even?" *Wiggle break + swap at ~12 min.* - **Challenge (7):** Try a look-alike card; talk about why it fools the computer. - **Share + discussion (10):** Pairs report what made guesses better, then the group discusses **why** a computer still guesses wrong sometimes: maybe it only saw one kind of example, or one group had way more examples than the other. Name one better example that would help, and talk about being fair — did the examples include everyone/everything? Model stays built.

Pair roles: Driver tests and shows cards, Navigator watches and reports right/wrong. Swap at the wiggle break. (*Trio: third keeps the example tally.*)

Differentiation: *Support* — focus on one group's examples. *Extension / fast-finisher* — compare two groups and say which needs more examples. *Reteach* — add one example at a time on the coach demo so the improvement is visible each time.

Exit check: Pair can name one reason a computer might guess wrong and one way better examples would help. **At-home (optional):** Explain to a family member why a computer needs lots of examples to learn well. ~5 min.

Lesson 30 — AI & Data: Design Challenge — AI Story Circle

Objective (I can...): I can design a little interactive build that uses recognition to change what happens. **CS concept:** Putting recognition together with events, conditionals, and counting · **Builds on:** Lessons 26–29 and the whole course. *Honest framing: kids use a ready-made recognizer — they do not train their own AI. Privacy: on-device camera, no logs, pictures never saved. Also serves as flex/catch-up — use today to finish and polish anything from the AI unit.*

AI Connection (2 min, read aloud): Today your build senses, decides, counts, and reacts to what a computer recognizes — every unit of the year working together. *Ask the group:* "Which idea from this year did your story use the most?"

Materials: LEGO CS & AI kit + device per pair with camera, plan mats, story props/cards. **Model already built** — reuse the AI & Data model (from L26); the 25 min is plan + code + play, not build-from-scratch.

60-minute flow - Warm-up (10): Story spark — pairs pitch a one-sentence story where showing a card or a pose changes what happens next. - **Concept + demo (8):** Challenge: build an interactive story where a recognized card (or pose) drives the model and the story. Coach shows a short plan connecting a recognition to a story beat. Restate privacy points. - **Build/Code (25):** Pairs plan and code an interactive story using the recognizer plus at least one if/then and, if they can, a count or a loop, on the built model. *Wiggle break + swap at ~12 min.* - **Challenge (7):** Add a second card that leads to a different ending. - **Share + tidy-up (10): AI story circle** — each pair performs their story while the coach records a short video clip for the parent note. **Photograph each build.**

Pair roles: Co-plan; Driver codes / Navigator directs the story and shows the cards; swap at the wiggle break. *(Trio rotates plan/code/perform.)*

Differentiation: *Support* — one recognition, one story beat, partial plan. *Extension / fast-finisher* — two recognitions with two endings and a count. *Reteach* — reuse their Lesson 28 program and add just one story reaction.

Exit check: Pair shows a working interactive story that uses recognition to change what happens, with a matching plan. **Progress layer:** Send home the **Unit 6 (course finale) parent note** ("your child can build a machine that senses, decides, counts, and can explain how a computer learns to sort from examples") with the **build photo** — the year's capstone. *Keep the framing honest: your child explored how AI works; training a classifier is next, in the 3–5 band.* **At-home (optional):** Draw a 3-picture interactive story; mark where the audience's card changes the path. ~5 min.

Standards Crosswalk

Aligned to the **CSTA K-12 CS Standards** — primarily **Algorithms & Programming (1A-AP)**, with connections to **Data & Analysis (1A-DA)** and **Impacts of Computing (1A-IC)**. These are the Level 1A (grades K-2) codes. This is a targeted map, not a claim of full coverage. *Maps to the Virginia CS Standards of Learning, grades K-2 (confirm exact SOL numbers).*

Unit (Lessons)	Skills taught	CSTA 1A codes targeted
1 • Basics (1-5)	Steps in order (sequencing), breaking a job into little steps (decomposition), saying the plan first	1A-AP-10 (model daily processes as algorithms/precise steps); 1A-AP-11 (decompose steps for a program); 1A-AP-08 (model a program's sequence of events)
2 • Events (6-10)	Triggers to start code; press, tap, and sensor inputs; more than one trigger	1A-AP-08 (programs with events/sequence); 1A-AP-10 (precise steps); 1A-AP-11 (decomposition)
3 • Loops (11-15)	Repeat loops, forever/stop, loop-in-a-loop (<i>stretch</i>), fixing a program, "hiding the mess"	1A-AP-10 (loops/repetition in algorithms); 1A-AP-14 (debug/fix errors); 1A-AP-08
4 • Conditionals (16-20)	If/then, true/not-true, if/then/else, sensor-cutoff decisions; comparing two solutions	1A-AP-10 (conditionals/decisions); 1A-AP-14 (debug); 1A-AP-12 (develop/compare plans — surfaced at design challenges)
5 • Variables (21-25)	Make/store a number (variable), counting up, a number in a decision, two numbers at once	1A-AP-09 (model the way programs store and manipulate data — a number in a box); 1A-AP-10
6 • AI & Data (26-30)	Sorting/classifying by a rule; examples are data; a recognizer as a trigger; why computers guess wrong (fairness)	1A-DA-05 (store, copy, and sort information to make it useful); 1A-DA-06 (collect and present data — example tallies); 1A-IC-16 (how people use computers / being fair); 1A-AP-10
Pair programming (every lesson)	Driver/Navigator roles, swapping, taking turns, working together	1A-AP-12 (work with others / follow and give instructions); collaboration across all lessons

Cross-cutting: Fix-It mini-tasks (L12, L14, L17, L19) reinforce **1A-AP-14**; the "two ways to do it" and design-challenge share-outs reinforce **1A-AP-12**; the AI-impacts prompts (L8, L13, L18, L25) and the L29 fairness discussion reinforce **1A-IC-16**; the sorting and example-tally work in Unit 6 reinforces **1A-DA-05 / 1A-DA-06**.

Honesty note for coaches and families: Throughout Unit 6, K-2 students **explore how AI works** — they sort examples, feed a coach-run demo, and use a ready-made recognizer to drive a build. They **do not build or train their own AI model**. Training a custom classifier is a grade 3-5 skill and lives in the Builders and Engineers bands. Keep that promise accurate on every flyer, parent note, and share-out.